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EXPLAIN HOW DATA IS STORED ON COMPUTERS

US 14944

Section B: Summative assessment

Task 1

REPORT ON COMPUTER DATA TYPES

1. INTRODUCTION

The purpose of this report is to demonstrate an understanding of computer data types, coding systems and data manipulation operations as observed in my organisation's computer system (HP 280 G3 desktop used for salon POS system, Pastel Accounting and Microsoft Office).

2. DATA TYPES WITH EXAMPLES (SO1, AC1)

Computer works with different data types to store different kinds of information.

a) Integer Data Type (Whole Numbers)

Definition: Integer is a whole number without decimal points. Used for counting.

Example:

- Quantity of shampoo bottles in stock = 50

- Invoice number = 1001

- Number of customers per day = 12

- Age = 25 years

- Student number = 2024001

Use: Counting, calculations that do not need decimals.

b) Real / Float / Double Data Type (Decimal Numbers)

Definition: Numbers with decimal points for precise values.

Example:

- Price = R899.99

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- Salary = R12,500.75
- Discount percentage = 15.5%
- Temperature = 36.6 degrees
- At salon POS, total sale = R450.50

Use: Prices, salaries, measurements.

c) Character and String / Text Data Type

Definition: Character is a single letter, digit or symbol ‘A’, ‘5’, ‘#’. String is group of characters “Hello”.

Example:

- Customer name: “Thandi Nkosi”
- Address: “123 Church Street, Pretoria”
- Service description: “Brazilian Treatment”
- Employee initial: ‘T’

Use: Names, addresses, descriptions.

d) Boolean / Logical Data Type

Definition: Only two values TRUE/FALSE or YES/NO.

Example:

- Is customer complaint resolved? TRUE
- Is product in stock? FALSE
- Is customer VIP? YES
- Paid? NO

Use: Decision making. Example in system: IF Paid = TRUE THEN Print Receipt ELSE Show Balance Due.

e) Date / Time Data Type

Definition: Stores calendar date and time.

Example:

- Complaint date: 05/08/2026
- Appointment time: 10:30 AM
- Expiry date of hair dye: 2026-12-31
- Date of Birth: 1998-05-12

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Use: Scheduling, sorting by date, calculating age, calculating overdue days.

f) Other Data Types (Extra marks)

Currency Data Type: Formatted as money with R symbol. Example R850.00

Memo / Object Data Type: Large text or objects. Example: Long complaint description, customer photo of damaged scalp, voice note complaint.

3. CATEGORIES OF CODING SYSTEMS USED IN A BUSINESS ENVIRONMENT (SO1, AC2)

Coding system is method to give short code to represent data for easy identification and classification.

i) Sequential Coding System

Definition: Numbers in sequence 001, 002, 003.

Example: Invoices numbered INV-001, INV-002, INV-003. Customer numbers 001, 002, 003. Complaint refs GR-001, GR-002, GR-003 at Greatermans Salon.

Advantage: Simple, easy to know next number.

ii) Block Coding System

Definition: Blocks of numbers reserved for specific category.

Example: Hair Products 100-199 (101=Shampoo, 102=Conditioner, 103=Hair Oil), Services 200-299 (201=Cut, 202=Colour, 203=Brazilian), Staff 300-399 (301=Manager, 302=Receptionist).

Advantage: Organises inventory by category.

iii) Group / Classification Coding System

Definition: Code shows class and sub-class, hierarchical.

Example: Employee code EMP-F-S-01 = Employee-Female-Salon-Staff number 01. Product code PROD-HR-DYE-RED = Product-Hair-Dye-Red colour. Customer code CUST-VIP-001 = Customer-VIP-001.

Advantage: Code tells you full details about item.

iv) Mnemonic / Significant Coding System

Definition: Abbreviation that is meaningful and easy to remember.

Example: SHAMP for Shampoo, COND for Conditioner, ACC for Accounting department, CUST for Customer, DISC for Discount.

Advantage: Staff remember easily, not just numbers.

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v) Barcode / QR Code and ASCII Coding System

Definition: Machine-readable code.

Example: Product barcode 6001234567890 on hair product scanned at till using barcode scanner for sales and stock. QR code on salon price list poster – customers scan with phone to see catalogue. Internal computer coding: ASCII code – computer converts letter 'A' to binary 01000001 to understand.

Advantage: Fast scanning, reduces errors.

4. CONCLUSION FOR TASK 1a

Different data types ensure accurate storage of different information and different coding systems make identification and searching fast in business environment.

TASK 1b (SO1, AC3) – HOW DATA MANIPULATION OPERATIONS ARE PERFORMED ON DATA TYPES – PRACTICAL DEMONSTRATION

Practical Demonstration Log

Venue: Greatermans Salon Office

System Used: Microsoft Excel 2016 and Pastel Accounting

Witness / Supervisor: Mr Mulaudzi

I demonstrated the following operations:

1. Insertion / Creation / Input Operation

Action: I opened Excel Customer Master file named Customer_Master.xlsx and inserted new customer record in Row 5:

ID=004 (Integer), Name=Lerato Dlamini (String), Phone=0721234567 (String), Date=07/08/2026 (Date), Amount=R1200 (Real/Float), Paid=FALSE (Boolean), Description="Overcharged R300" (Memo).

Result: New record added successfully. Supervisor observed.

2. Retrieval / Searching / Reading Operation

Action: I used Excel FILTER to find Customer "Thandi Nkosi". I used formula =VLOOKUP("GR-001",A2:G10,3,FALSE) to retrieve her complaint details from Complaint Master file. I demonstrated how Pastel searches indexed file using Customer ID.

Result: Retrieved complaint GR-001 details within 2 seconds.

3. Updating / Modification / Editing Operation

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Action: I changed Lerato's Amount from R1200 to R900 after correcting overcharge error (Real data manipulation). Updated Paid field from FALSE to TRUE after refund processed (Boolean manipulation). In Pastel, edited product price from R899.99 to R950.00.

Result: Updated successfully. Old value overwritten.

4. Deletion / Purging Operation

Action: I deleted duplicate entry Row 15 which was duplicate invoice INV-002. Before deletion I explained need to make backup copy.

Result: Duplicate removed. File size reduced.

5. Sorting Operation

Action: I sorted customer names A-Z (String sorting) – Order became Dlamini, Molefe, Nkosi. I sorted sales amount Largest to Smallest (Real number sorting) – R1200, R850, R250.

Result: Data arranged in order for easy reporting.

6. Calculation / Arithmetic Operators (+, -, *, /, %)

Action: In Excel I calculated:

Total = Quantity * Price = 2 * R450.50 = R901.00

VAT = Total * 15% = R901.00 * 0.15 = R135.15

Grand Total = Total + VAT = R1036.15

Balance = Grand Total – Amount Paid

Demonstrated numeric data types operation using + - * / %

7. Comparison Operators (>, <, =, >=, <=, <>)

Action: I used formula =IF(Amount>1000,"High Value Customer"," Low Value Customer") to classify customers. Filtered complaints where Severity="High".

Example: 1200 > 1000 = TRUE, so Lerato is High Value.

8. Logical Operators (AND, OR, NOT)

Action: I used formula =IF(AND(Paid=FALSE, Amount>1000), "Follow Up Urgently", "OK") for credit control. Used OR to show customers who either have High severity OR Amount >1000.

Example: For Lerato, Paid=FALSE AND Amount=1200>1000 = TRUE, so "Follow Up Urgently".

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TASK 2 (SO2, AC1, AC2, AC3) – DESCRIBE COMPUTER DATA STRUCTURES [50 MARKS]

REPORT ON COMPUTER DATA STRUCTURES, COMPUTER FILES AND COMPUTER DATABASES

1. INTRODUCTION

This report describes data structures, file types and database types used to store data efficiently in computers at Greatermans Salon and College PC Lab.

2. TYPES OF COMPUTER DATA STRUCTURES WITH EXAMPLES (SO2, AC1)
    a) Array Data Structure

Definition: Array is a fixed-size collection of elements of the same data type stored in contiguous (next to each other) memory locations. Each element is accessed by an index number starting from 0.

Example: At Greatermans Salon, daily sales for 7 days can be stored as Array: Sales[0]=R1000 Monday, Sales[1]=R1200 Tuesday, Sales[2]=R900 Wednesday, Sales[3]=R1500 Thursday, Sales[4]=R2000 Friday, Sales[5]=R2500 Saturday, Sales[6]=R800 Sunday. Student marks array: [65,70,80,55,90] for 5 modules.

Advantages: Fast access using index, simple. Disadvantages: Fixed size cannot grow, wastes memory if not full.

    b) Linked List Data Structure

Definition: Linked List is a dynamic data structure where each element called a node contains data and a pointer that points to next node. No fixed size.

Example: Customer waiting list at salon: Node1 Thandi Nkosi -> points to Node2 John Molefe -> points to Node3 Lerato Dlamini -> points to NULL (end). If new customer arrives between Thandi and John, we only change pointers.

Advantages: Dynamic size can grow/shrink, easy insertion/deletion. Disadvantages: Slower to search because must follow pointers.

    c) Stack Data Structure (LIFO – Last In First Out)

Definition: Stack works like pile of plates. You add (PUSH) on top and remove (POP) from top. Last item added is first to be removed. LIFO.

Example:

1. Browser history – last website visited is first to go back when click back button.

2. Undo operation in Word – last action is first undone.

3. Stack of clean towels at salon – last towel placed on top is first used.

4. Stack of invoices on manager desk.

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Operations: PUSH (add), POP (remove), PEEK (view top).

d) Queue Data Structure (FIFO – First In First Out)

Definition: Queue works like queue at shop till. First person to enter queue is first served. FIFO.

Example:

1. Customer complaint queue – First complaint GR-001 handled before GR-002 and GR-003.

2. Print queue – first document sent to printer prints first.

3. Call centre queue at NewYou Salon Durban – first caller answered first.

Operations: ENQUEUE (add at rear), DEQUEUE (remove from front).

e) Tree / Hierarchical Data Structure

Definition: Tree is hierarchical structure with root at top and parent-child relationship. Each node can have many children.

Example:

1. Company organogram: Root=Owner Mr Mulaudzi, Children=Salon Manager and Receptionist, Grandchildren=Stylists reporting to Manager.

2. File system: C: Drive -> Documents -> POE folder -> Complaint Handling Folder -> Task1.docx

3. Decision tree for complaint handling.

Advantage: Represents hierarchy clearly, fast searching.

f) Record / Structure / Table Data Structure

Definition: Record is collection of different data types about one entity in one row.

Example: Customer Record = {ID: Integer 001, Name: String “Thandi Nkosi”, Phone: String “0721234567”, Amount: Real 850.00, Paid: Boolean FALSE, ComplaintDate: Date 05/08/2026, Description: Memo “Hair dye burning scalp”}

3. TYPES OF COMPUTER FILES WITH EXAMPLES (SO2, AC2)

i) Master File

Definition: Permanent file that holds main, up-to-date and complete information about business entities. Reference file updated periodically with transaction files.

Example: Customer Master File at Greatermans Salon contains all 500 registered customers with names, phones, balances. Product Master File contains all hair products with current stock and price. Employee Master File.

Nature: Static, permanent, very important. Needs backup.

ii) Transaction File

Definition: Temporary file that holds daily transactions or changes that will update master file.

Example: Daily Sales Transaction file for 23/08/2026 contains 20 sales made today. Daily Complaint Transaction file contains 3 new complaints today before merging to master complaint file.

Nature: Dynamic, temporary. At end of day merged into master and cleared.

iii) Serial File

Definition: Records stored in order they arrive, no sorting. Chronological order.

Example: Complaint forms dropped in complaint box as they arrive. Log of phone calls 08:00, 08:05, 08:10. CCTV footage log.

Access: Must read from start to find item (serial access). Slow.

iv) Sequential File

Definition: Records sorted by key field such as ID or surname alphabetical. Must be accessed sequentially from start.

Example: Employee file sorted by Employee Number 001,002,003. Customer file sorted by surname A-Z: Dlamini, Molefe, Nkosi. Invoice file sorted by Invoice Number.

Access: Must read one by one but in sorted order. Good for reports.

v) Indexed File / Indexed Sequential File (ISAM)

Definition: File has index table like index at back of book. Index stores key and address of record, so system can jump directly to record without reading all.

Example: Customer file indexed on Customer ID – to find Customer 005 with complaint GR-002, system checks index and jumps directly. Like dictionary. In Pastel and salon CRM, search customer by cell number using index – fast.

Access: Direct via index plus sequential. Fast searching.

vi) Direct / Random / Relative File

Definition: Records stored with address calculated by formula (hashing). Can access any record directly without reading others.

Example: RAM memory – CPU goes directly to memory address 0x1234. Appointment system – go directly to time slot 10:00 without checking 09:00. Booking number 10 stored at location 10 on disk.

Access: Fastest, instant direct access. Used for real-time systems.

4. TYPES OF COMPUTER DATABASES WITH EXAMPLES (SO2, AC3)

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i) Hierarchical Database

Definition: Organised like upside-down tree. One parent can have many children, but one child can have only ONE parent. One-to-Many (1:M) relationship.

Structure: Root at top, branches below.

Example: Organisation chart – Owner is parent of Manager, Manager is parent of Stylists. File system – C: drive is parent of Documents folder. Old system IBM IMS.

Business Example: At Greatermans Salon, Salon -> Customers -> Appointments. Customer Thandi cannot belong to two salons at same time in this model.

Advantages: Very fast for fixed routes, simple. Disadvantages: Inflexible, child cannot have two parents, delete parent deletes all children.

ii) Network Database

Definition: Like hierarchical but more flexible – child can have many parents. Many-to-Many (M:M) relationship. Uses pointers to link records.

Structure: Graph / Net structure.

Example: Student database – One Student can register for MANY Subjects, and one Subject has MANY Students. At Greatermans Salon: One Customer can be served by many Stylists over time, and one Stylist serves many Customers. One Product can be in many categories.

System: IDMS, IDS.

Advantages: More flexible than hierarchical, represents real world better. Disadvantages: Complex to design and maintain.

iii) Relational Database

Definition: Most modern and most used today. Data stored in TABLES called Relations. Each table has Rows (Records) and Columns (Fields). Tables linked by common field called Primary Key (PK) and Foreign Key (FK). Uses SQL (Structured Query Language) to query.

Structure: Tables with keys.

Example Design for Greatermans Salon:

Table 1: Customers (CustomerID PK, Name, Phone)

Row1: 001, Thandi Nkosi, 0721234567

Row2: 002, John Molefe, 0731234567

Table 2: Complaints (ComplaintID PK, CustomerID FK, Description, Amount, Severity, Date)

Row1: GR-001, 001, Hair dye burning, 850.00, High, 05/08/2026

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Row2: GR-002, 002, Waiting time, 0.00, Medium, 06/08/2026

Link: CustomerID links both tables. One Customer has Many Complaints.

SQL Query Example: SELECT * FROM Complaints WHERE Severity = 'High';

Systems: Microsoft Access (used for my POE), MySQL, SQL Server, Oracle, Pastel database uses relational model.

Advantages: Very flexible, easy to use, easy to query with SQL, easy to add new tables, avoids duplication, many-to-many possible via junction table. Most used in all businesses today.

Disadvantages: Slightly slower than hierarchical for very large fixed structures, but still best.

DIFFERENCE COMPARISON

Feature	Hierarchical	Network	Relational
Structure	Tree structure ,upside down	Graph/ net structure	Tables with rows and columns
Relationship	One to many	Many to many	Many to many via primary key and foreign key
Parent rule	One parent	Many parents	No parent/child, linked with keys
Flexibility	Low	Medium	High
Query method	Navigational, must follow parent to child	Navigation ,using pointers	SQL
Example	File system on computer	Student-Subject enrollment	Customers and complaints table
Used today	Old system, file storage	Rare, complex system	Mixture of old and new system

5. CONCLUSION

In conclusion, data structures like Array, Linked List, Stack, Queue and Tree help organise data in memory for efficient processing. Different file types like Master, Transaction, Serial, Sequential, Indexed and Direct help store data on disk with different access methods. Different databases like Hierarchical, Network and Relational help manage large business data. For Greatermans Salon, relational database is best because it is flexible, easy to query and easy to manage customer complaints and sales data.